

SULPHUR SPRINGS, TX, USA

WMO: 722362

Lat: 33.167N

Lon: 95.617W

Elev: 149

StdP: 99.55

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 93937

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.2	-2.2	-11.6	1.4	-0.4	-9.1	1.8	1.6	12.3	9.8	11.4	9.3	4.1	0	0.450

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.9	37.7	23.0	36.8	23.0	35.3	23.2	25.6	31.8	25.2	31.2	24.8	30.8	3.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.1	19.4	27.4	23.7	18.9	27.3	23.2	18.3	27.1	79.3	31.5	77.5	31.2	75.8	30.8	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.3	8.8	7.9	-8.0	39.3	1.7	1.4	-9.2	40.3	-10.2	41.1	-11.2	41.9	-12.4	42.9		
			-8.9	26.2	1.7	0.7	-10.1	26.7	-11.1	27.1	-12.0	27.5	-13.2	28.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.9	7.8	9.9	14.5	18.2	22.7	27.5	29.0	29.3	25.7	19.4	13.8	9.0
	DBStd	8.74	5.40	5.79	5.22	4.16	3.59	2.18	2.31	2.59	3.23	4.64	5.17	5.26
	HDD10.0	303	107	68	20	2	0	0	0	0	0	1	22	83
	HDD18.3	1256	328	241	138	53	9	0	0	0	0	43	153	291
	CDD10.0	3568	38	64	159	248	394	524	589	599	471	293	135	53
	CDD18.3	1481	1	4	19	49	145	274	331	341	221	77	16	3
	CDH23.3	16493	6	19	108	353	1255	3137	4147	4355	2329	692	79	14
	CDH26.7	7932	0	1	9	60	421	1499	2213	2406	1099	215	9	0
Wind	WSAvg	3.4	3.7	3.9	4.1	4.0	3.5	3.3	3.0	2.7	2.7	3.1	3.5	3.5
Precipitation	PrecAvg	1167	85	91	110	96	117	103	80	62	86	123	105	102
	PrecMax	1704	182	221	258	200	290	369	223	148	226	400	331	211
	PrecMin	569	1	13	12	34	32	15	2	2	21	7	17	11
	PrecStd	289	45	57	58	46	66	78	68	38	49	95	76	56
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.1	25.3	27.8	30.4	33.5	37.2	38.9	40.1	37.4	32.9	27.9	25.1
		MCWB	15.0	17.4	17.8	20.0	22.5	22.8	23.0	22.6	21.4	21.6	19.6	17.8
	2%	DB	22.0	23.2	26.2	28.0	32.1	35.5	37.4	38.0	35.5	31.1	25.7	22.6
		MCWB	15.7	16.7	17.9	19.9	22.3	22.9	22.8	23.2	22.5	21.8	18.5	17.2
	5%	DB	19.5	21.4	24.2	27.1	30.7	34.1	36.2	37.2	33.9	29.2	23.7	20.6
		MCWB	14.5	16.3	16.9	19.3	22.0	23.2	23.3	23.3	22.5	20.2	17.9	16.1
	10%	DB	17.3	19.6	22.6	25.5	29.1	32.9	34.9	35.8	32.6	27.4	22.2	18.0
		MCWB	12.5	15.2	16.6	18.4	21.5	23.3	23.6	23.3	22.0	19.3	17.4	14.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.8	19.9	20.5	22.9	24.6	26.1	26.1	26.2	25.3	24.1	21.8	20.2
		MCDB	21.7	22.7	25.0	27.2	29.4	31.6	32.4	33.2	30.7	28.7	25.9	22.6
	2%	WB	17.2	18.9	19.5	21.6	23.7	25.3	25.5	25.6	24.7	23.4	20.6	18.7
		MCDB	19.8	22.0	23.6	26.3	28.9	30.6	31.7	32.3	30.0	27.9	23.4	21.7
	5%	WB	15.6	17.6	18.7	20.7	23.1	24.8	25.1	25.2	24.1	22.6	19.5	17.0
		MCDB	18.5	20.1	22.5	25.0	28.4	30.2	31.3	31.6	29.4	27.0	22.6	19.5
	10%	WB	12.9	15.7	17.8	19.9	22.5	24.3	24.7	24.7	23.7	21.6	18.0	15.0
		MCDB	16.4	19.0	21.5	24.1	27.6	29.9	30.8	31.0	29.0	25.8	21.4	17.7
Mean Daily Temperature Range	5% DB	MDBR	10.3	10.1	10.7	10.9	10.2	10.1	10.4	10.9	11.3	11.6	11.0	10.1
		MCDBR	13.6	12.4	11.9	11.7	11.1	11.3	11.8	12.4	12.6	12.4	11.7	13.4
		MCWBR	8.7	7.9	6.2	5.6	4.1	2.9	2.6	2.7	3.6	4.7	6.3	9.0
	5% WB	MCDBR	11.8	10.8	9.8	9.8	9.9	9.5	9.9	10.4	10.2	10.0	9.8	11.6
		MCWBR	9.4	8.1	6.3	6.0	4.3	3.3	3.0	2.9	3.4	5.1	7.6	9.2
Clear-Sky Solar Irradiance	taub	0.316	0.331	0.358	0.392	0.427	0.439	0.450	0.448	0.401	0.357	0.340	0.320	
	taud	2.489	2.460	2.404	2.320	2.283	2.288	2.294	2.286	2.372	2.472	2.467	2.486	
	Ebn at Noon	902	922	919	897	864	850	838	835	863	881	864	875	
	Edh at Noon	86	99	113	128	134	133	132	131	115	96	88	83	
All-Sky Solar Radiation	RadAvg	2.73	3.13	4.29	5.27	5.73	6.39	6.50	6.03	5.05	3.99	2.92	2.40	
	RadStd	0.26	0.48	0.35	0.36	0.50	0.62	0.42	0.43	0.47	0.57	0.37	0.31	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page